FNB APP OF THE YEAR HACKATHON 2026 | ACCESS TO HIGHER EDUCATION

# ThutoVest

## From “I want to study” to “I can graduate.”

A personal plan, early warnings and verified sponsors who pay the university directly. Free for students.


TEAM THUTO

**Samalani Maluwa**

**Reshoketswe Makhura**

**Tebogo Sasa**


**Scan to try it**

thutovest.vercel.app

*Thuto* means learning in Setswana and Sesotho.
*Vest* means investing in a student.

IN THIS PRESENTATION

# The questions this deck answers

- **Why** — Why does this need to exist?
- **Who** — Who is it for?
- **What** — What is ThutoVest?
- **How** — How does it work, for students and for sponsors?
- **How** — How does it keep students on track after funding?
- **How** — How is it fair, and how are children protected?
- **How** — How does it think, and how is it kept safe?
- **How** — How is it built, and how does it make money?
- **What** — What is real today, and what comes next?
- **Try** — How do I try it in three minutes?

WHY

# Matriculants plan their future alone

The challenge: help young South Africans choose a path, fund it, apply and stay supported until graduation.

**893,847**
NSFAS funding applications for 2026

**218,000**
still incomplete because documents were missing

### Scattered information
A plan pieced together from NSFAS, university sites and WhatsApp groups.

### Missed deadlines
Nothing warns them in time, so applications and funding slip away.

### The missing middle
Households on R350,000 to R600,000 earn too much for NSFAS, but can’t pay fees.

### Bursary scams
Fake offers ask for “admin fees”, and students can’t tell real from fake.

### Funded, then dropped
Marks slip quietly. Students hear about it when the funding is already gone.

Sources: NSFAS board chair, reported by TimesLive and EWN on 6 January 2026. Missing middle: households earning R350,000 to R600,000 a year, above the NSFAS threshold. Scams: Daily Maverick, 8 April 2025.

WHO

# Built for three people around one student

## Students

Grade 11 to university

- Tatenda, Grade 12, 17, Gauteng. Wants Computing and IT, APS 32, a R92,400 gap.
- Nandi, first-year at UJ, funded. Wants to keep her bursary.
- Free, on any phone, with WhatsApp for low data.

## Parents and guardians

For learners under 18

- Give consent before any sponsor can see the child.
- Co-sign the bursary agreement.
- Today they do this on the student's phone. Next: their own link, with copies of offers and warnings.
- Nomsa, Tatenda's mother, in the demo.

## Sponsors

Companies, trusts and non-profits

- Find verified, ranked students who fit their fund.
- Set their own terms and pay the institution directly.
- See the progress of students they fund.

WHAT

# ThutoVest is one place to plan, fund and finish a degree

It turns a student's answers into a personal plan with one next best action, connects them with verified sponsors, and keeps support going once they are studying.

## Plan

A personal pathway from choosing a course to first year: course matches, deadlines, documents and Risk Radar warnings.

## Fund

The funding gap in rands, NSFAS and other aid, and verified sponsors who pay the institution, never the student.

## Stay

Marks analysed every time they are added, advice, tutor requests, and fair early warnings before funding is at risk.

**66**

qualifications in 8 fields

**76**

public universities and TVET colleges, with official links

**2 sec**

for an offer to reach the student's phone

**Free for students, always. Sponsors pay a small admin fee on awards paid through the platform.**

HOW IT WORKS

# Seven stops from matric to graduation

At every stop the app shows one next best action, so the student always knows what to do next.

1

### Explore

Courses that fit the APS, field and budget. Change the APS and every match is re-scored.

2

### Apply

Official apply links for 26 universities and 50 TVET colleges, with alerts before closing dates.

3

### Funding

The gap in rands, and which aid can close it: NSFAS, ISFAP, Funza Lushaka and more.

4

### Offers

A verified sponsor finds the anonymised profile and sends an offer, never above the gap.

5

### Register

Sign the agreement (a guardian co-signs if under 18) and share documents in one tap.

6

### Start

The sponsor approves and pays the institution directly.

7

### Stay

Marks are analysed, slips are caught early, and a tutor is one tap away.

**Tatenda's example:** APS 32, Computing and IT. The plan shows a R92,400 gap. Ridgeline offers R60,000 for tuition and accommodation, Nomsa co-signs, and R32,400 is left for a second bursary. Other sponsors only see what is left.

HOW IT WORKS FOR STUDENTS

# Everything a student needs, in one plan

### Study Explorer
66 qualifications in 8 fields, every field with at least six routes, including diplomas and higher certificates.

### Risk Radar
Early warnings for deadlines, missing documents and money gaps, each with its reason.

### Funding Gap Solver
NSFAS, ISFAP, Funza Lushaka, university aid and loans in one view, with what-if scenarios.

### Universities and colleges
All 26 public universities, all 50 public TVET colleges and the DHET private college register, with official links.

### Check a place before you pay
NSFAS 2026 accommodation rules and scam signs, before a student pays a deposit.

### Document vault
Automatic checks for type, readability and editing. Shared with a sponsor in one tap.

### Marks and progress
Every new set of marks is analysed, with advice and a tutor request for funded university students.

### Talk to ThutoVest
A WhatsApp bot for students low on data: deadlines, offers and documents.

HOW IT WORKS FOR SPONSORS

# Fund the right students, with less admin and no fraud

- Register the organisation. Identity, representative and bank checks before any fund goes live.
- Publish funds with their own terms: what the bursary pays for, a minimum pass mark, work-back years, and whether it must be the only bursary.
- Candidates ranked by fit and by what this sponsor tends to fund, with the reasons shown.
- Offers, document review, funding approval and payment to the institution, never to the student.
- A forecast of the date each fund runs out, and a skills-development spend estimate.

## Student progress

For every student they fund, sponsors see each module mark as soon as it is added, flagged when a student slips below the pass mark.

**Zanele**, first-year at NMU: 56% against a 60% pass mark, Data Structures and Algorithms at 44%, and a tutor request waiting.

**One click sends a fair early warning, or arranges the tutor.**

HOW A BURSARY FLOWS

# From first match to payment, every step checked on the server

**1**

### Match

The student shares an anonymised, verified profile, or the sponsor finds them.

**2**

### Offer

The sponsor offers an amount. The server refuses anything above the gap left, or a third bursary.

**3**

### Sign

The student accepts and signs in the app. Under 18? The guardian co-signs.

**4**

### Documents

Required documents are shared and checked automatically. A child's offer always needs guardian documents.

**5**

### Approve

Payment stays locked until every required document is approved.

**6**

### Pay

The sponsor pays the institution directly. The student's gap drops for everyone.

**Safety at every step: a message asking for an "admin fee" is blocked and the sponsor suspended. Real sponsors never ask students for money.**

HOW IT KEEPS STUDENTS ON TRACK

# Every time marks are added, the app reads and analyses them

Funding is only half the job. ThutoVest catches a slip while there is still time to fix it.

| | |
|---|---|
| **1 Add marks** | Type them in, or read them from a report (simulated in this demo). University students add module marks and credits. |
| **2 Analyse** | Average, change since last time, APS from NSC levels, slipping subjects and the bursary's pass mark. |
| **3 Advise** | On track, keep an eye on it, or at risk, with reasons, the courses now out of reach, and free help. |
| **4 Ask for a tutor** | The assistant offers to send a tutor request to the sponsor. One tap to approve. Contact details are removed. |
| **5 Early warning** | The sponsor sends a warning the platform words fairly: the requirement, no decision yet, and help. |

### The warning a student receives

"Your latest average is 56% (Semester 2 tests, 2026), below the 60% this bursary requires. **This is not a decision to stop your funding.** If your results don't satisfy the bursary's requirement at the next review, you could be at risk of losing this funding. Let's make a plan together: reply here, and ask us for a tutor if it would help."

**Only the sponsor funding a student sees their marks. A warning note that asks for money is refused.**

HOW IT SUSTAINS A STUDENT

# From admission to graduation, protecting the funding every year

Students should never lose a bursary out of nowhere. ThutoVest turns a year-end shock into early help.

**1**

### Admission

Registered, and the sponsor has paid the institution. The bursary's conditions, such as a 60% pass mark, are clear from day one.

**2**

### Every test

The student adds marks. The app compares them with last time and with the pass mark, and catches a slip before results are final.

**3**

### Early help

The assistant offers to ask the sponsor for a tutor. The student approves with one tap, and the sponsor arranges it.

**4**

### Early warning

If marks keep slipping, the sponsor sends a fair warning: the requirement, no decision yet, and an offer of help.

**5**

### Every year

Reminders to renew funding and re-register, and a second bursary if the gap grows, within the two-bursary limit.

**The cycle repeats every year until graduation.**

### Without ThutoVest

Marks slip quietly all year. A letter after final results says the bursary has stopped, when it is too late to fix.

### With ThutoVest

The slip shows up after the first test. The student gets a tutor, the sponsor gets a plan, and the funding is kept.

HOW IT IS FAIR

# Fair funding rules, and children protected by law

## Fair funding

- A student holds at most two sponsor bursaries, so one can pay tuition and another the rest.
- Sponsors see the gap left after other bursaries, and no offer or acceptance can go above it.
- An “only bursary” can’t be combined with another.
- Disability-only bursaries match students who say they have a disability. Sponsors see yes or no, never details.
- Money only ever goes to the institution.

## Under-18s (POPIA sections 34 and 35)

- Hidden from every sponsor until a parent or guardian consents. The wording and time are recorded.
- Every offer to a child requires the guardian’s consent form, certified ID and proof of guardianship.
- Proof is a birth certificate, or a sworn affidavit if the guardian isn’t a parent.
- The guardian co-signs the bursary agreement.

All enforced on the server. A demonstration build: these legal controls have not been reviewed by a lawyer.

HOW IT THINKS

# The app predicts problems before they happen

Explainable rules built on each user's own data. Every alert has a "Why?" that shows the evidence.

| IT LEARNS FROM | SO IT CAN |
|---|---|
| **APS, field, budget and province** | Re-score every course match the moment the APS changes |
| **Marks after every test or term** | Spot a slipping subject, recalculate the APS and warn before a bursary is at risk |
| **The student's pace and activity rhythm** | Predict who will fall behind, and nudge before they do |
| **Costs, aid and deadlines** | Show the funding gap month by month, with what-if scenarios |
| **A sponsor's past decisions** | Rank the students that sponsor is most likely to fund |
| **Fund budget and award pace** | Forecast the date each fund runs out |
| **Message text and document contents** | Block fee requests, and flag the wrong type, unreadable or edited files |

Honest note: these are explainable rules, not machine learning yet. Machine learning comes once there is real outcome data to train on.

HOW IT IS KEPT SAFE

# Security built in, and tested against the live app

**95**
automated checks pass on every change

**38 of 38**
live attack probes refused

**8 of 8**
database tables with row-level security

- Every action is checked on the server with the same rules code as the app, so browser and server never disagree.
- Fee and bank-detail requests are blocked and the sender suspended. Sponsors can’t verify themselves.
- Uploads are checked by their real file contents. WhatsApp refuses ID numbers, OTPs, bank details and files.
- Marks go only to the sponsor funding the student. Each demo account resets only its own records.
- Signed webhooks, strict Content Security Policy, HSTS, no secrets in code or history. Demo logins are locked.

HOW IT IS BUILT

# One set of rules, on the phone and on the server

## On the phone

- Web app: React 19, TypeScript, Vite and Tailwind
- Works on any phone or laptop
- WhatsApp: Talk to ThutoVest for low data

## In between

- Vercel serves the app with strict security headers
- Twilio forwards WhatsApp messages as signed webhooks
- Data and logins over HTTPS

## On the server (Supabase)

- Auth: real logins with bcrypt-hashed passwords
- Platform function: the app's own rules check every action
- Postgres with row-level security on every table
- Realtime: changes reach open screens in about 2 seconds

Without the backend settings, the same app still runs fully in the browser, so the demo never depends on one service.

WHAT WE BUILT

# Built during the hackathon, 23 to 26 October

The student planning core existed before the event. These were built and posted in our Sonke progress updates.

### Live backend

Real logins, server-side rules, row-level security and live sync.

### WhatsApp bot

Live on real phones through Twilio, opened from the app.

### Real institution data

All 26 universities, 50 TVET colleges, the DHET register and NSFAS residence rules.

### Fair funding rules

Bursary terms, at most two bursaries, and no offer above the gap left.

### Children protected

Guardian consent recorded, and guardian documents on every offer to an under-18.

### Marks and progress

Analysis on every upload, advice, tutor requests and fair early warnings.

### More study routes

66 qualifications, at least six in every field, including diplomas.

### Security hardening

38 of 38 live attacks refused, 95 automated checks, demo logins locked.

HOW IT MAKES MONEY

# Free for students, paid for by the funding it unlocks

## How it earns

- Free for students, always.
- A small admin fee on awards paid through the platform, charged to the sponsor’s award, never to the student.
- Impact and skills-development reports for sponsors.
- Cohort insights for schools and institutions.

## What comes next

- Real identity and bank verification partners
- USSD menus for phones with no data
- NSFAS status and institution confirmations
- Email and SMS alerts, and personal WhatsApp updates
- Forecasts trained on real outcomes
- Real sign-up on the server for new students and sponsors

WHAT IS REAL

# What is real today, and what is simulated

## Real

- Live app with real logins and live sync between devices
- Server-side rules and row-level security
- WhatsApp bot on real phones (Twilio sandbox)
- Official university and college data, with sources and dates
- Document type and tamper checks, scam blocking
- Bursary, child-safety, marks and progress rules, enforced on the server

## Simulated, and labelled in the app

- Identity, guardian and bank verification
- Payments to institutions and institution confirmations
- Sponsors, students and awards: fictional
- Fees, APS guides and some deadlines: demo figures
- Reading marks from a report: the student checks them
- The Navigator chat is scripted; WhatsApp shows the demo student

A demonstration build. The POPIA and children’s controls were designed by the team from the Act and have not been reviewed by a lawyer.

QUESTIONS JUDGES ASK

# What, why and how, answered

### How is this different from NSFAS or a bursary website?

It joins the whole journey: plan, apply, fund and stay on track, with one next action at every step. Sponsors pay institutions, not students.

### Why would a sponsor use it?

Verified, ranked candidates, their own terms enforced, progress of the students they fund, and impact reports. Less admin, no fraud.

### How do you stop scams?

Messages are screened. Fee and bank requests are blocked and the sender suspended. Money only goes to institutions.

### How do you protect children?

Under-18s stay hidden until a guardian consents. Guardian documents and a co-signature on every offer.

### How do you stop students losing funding?

Marks are read after every test. The assistant offers a tutor request, and the sponsor can warn early. No surprises at year end.

### How does a parent see what is happening?

Today on the student's phone: consent and co-signing. Next: a guardian link with copies of offers and warnings.

### What if a student has no data?

WhatsApp works today. USSD menus are next.

### Is the intelligence real AI?

Explainable rules from each user's own data, with the evidence shown. Machine learning once real outcomes exist.

TRY IT

# Try it in three minutes at thutovest.vercel.app

Use two screens: one tab signed in as a student and one as the sponsor, side by side on the same computer (each tab keeps its own login), or two phones. Every password is Demo#2026.

| | |
|---|---|
| **Tatenda, Grade 12, 17** | **tatenda@thutovest.demo** |
| **One login per judge** | **judge1 to judge5@thutovest.demo** |
| **Nandi, funded first-year at UJ** | **nandi@thutovest.demo** |
| **Ridgeline Technologies (sponsor)** | **funding@ridgeline.demo** |

WhatsApp: send **join salmon-castle** to **+1 415 523 8886**, then **hi**.

## The three-minute demo

1. Tatenda: approve Share my profile with Ridgeline.
2. Ridgeline: open Candidates and make an offer.
3. Tatenda: review, sign and accept. Nomsa co-signs.
4. Ridgeline: approve documents, then pay the institution.
5. Nandi: Marks and progress, Use a sample report, Save and analyse. On the dashboard, approve the assistant's tutor request.
6. Ridgeline: Student progress, send Zanele an early warning.

TEAM THUTO

# Every student deserves a plan, a fair chance at funding, and support until graduation.

**Samalani Maluwa**

**Reshoketswe Makhura**

**Tebogo Sasa**

**Thank you. Try it now.**

thutovest.vercel.app

**thutovest.vercel.app | github.com/SamkiDev/thutovest**